Memory that remembers. Beliefs that can change. Boundaries that hold.

Omega Hive × ASI

From replaceable modules to private memory and guarded native revision

M2 · C1 + C2 COMPLETE

REAL HIVE · DETTA · PLN

Architecture and evidence • 16 October 2026 • supported local scope

ELI16: a notebook, an evidence ledger, a calculator

An agent needs to remember what happened without treating every report as a new fact.

C1 · THE NOTEBOOK

Keep the original story

Record events in order, including who they came from. Replaying the same event does not add a new entry.

C2 · THE EVIDENCE LEDGER

Keep disagreement visible

Track which observations support a belief. A copied report does not become independent evidence.

PINNED PLN · THE CALCULATOR

Update only when justified

Use the existing formula for an explicitly allowed pair. Save the decision and the evidence behind it.

What works

Real Hive + local storage + native revision survive tested process restarts.

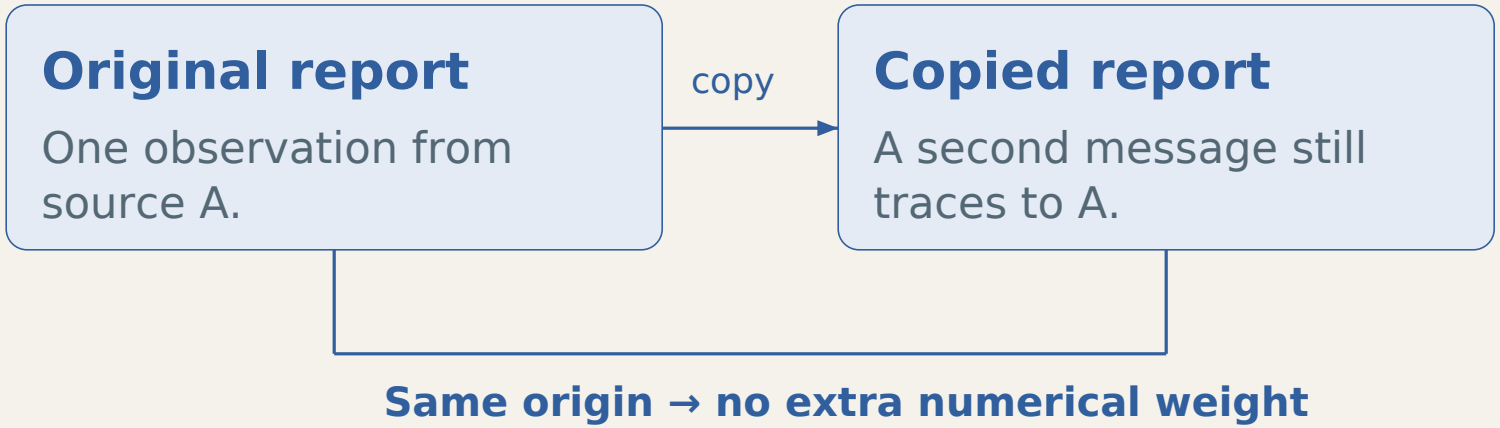
What this does not prove

That reports are true, sources are independent, or the system is production-ready.

Why it matters: we can inspect, retry and replace parts without silently changing their meaning.

ELI16: a copied report is not a new source

Imagine deciding whether a sensor reading is reliable. Keep the source trail, not just the message count.



Contrary observation

Source B can disagree; retain it.

Store all three observations

Remembering a report does not mean believing it.

Combine only an eligible pair

Explicit independence assumption + lineage guards + native formula.

A retry is not a new event. A copy is not a new source. A disagreement is not erased.

Fifteen modules; four responsibility groups

A module owns a responsibility. It need not be a separate process or repository.

CLIENT / COGNITION

- Episodic memory
- Belief revision
- PLN kernel
- ECAN / attention
- Inference control

APPLICATION PORTS

- AtomStorage
- Relations · future
- Search projection · future
- Agent messaging
- Delegated execution

CONTROL & POLICY

- Registry
- Compatibility resolver
- Authorization

Local test policy
and static descriptors

PROVIDER ADAPTERS

DA storage

Real local DeTTa

Consensus / finality

M1 historical; M2 NOT_RUN

SPECIFIED / FUTURE IMPLEMENTED LOCAL REAL PROVIDER

Narrow RevisionMath ≠ general PLN kernel

Data responsibilities: bytes → references → history

Persistence, identity, structure, discovery and provenance have different owners.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / STATUS
DA storage	Persist opaque bytes put · get · availability	Payloads + shares REAL LOCAL PROVIDER
AtomStorage	Scoped immutable envelopes create · get · getMany · availability	Reference + replay metadata IMPLEMENTED
Relations	Represent ordered structural roles assertRelation · getRelation	Immutable relation records SPECIFIED
Search projection	Find refs at a known watermark query	Rebuildable index SPECIFIED
Episodic memory C1	Record / append / read ordered events pendingEvidence · acknowledgeEvidence	Private history + delivery intents IMPLEMENTED · M2

DA persists bytes. Only the application owners interpret events, claims or beliefs.

Cognitive responsibilities: history ≠ truth ≠ attention

M2 implements belief revision. General rule execution and attention remain future work.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / BOUNDARY
Belief revision C2	Admit evidence; read immutable snapshots submitEvidence · readBelief	Evidence, supports, snapshots, attempts IMPLEMENTED · no agenda
PLN kernel	Apply a rule to explicit premises profiles · applyRule	No mutable domain state SPECIFIED · not RevisionMath
ECAN / attention	Allocate, select and decay attention submitStimulus · select · advance	Balances + logical time SPECIFIED · no PLN dependency
Inference control	Schedule bounded reasoning work step	Receipts + checkpoint SPECIFIED · no truth ownership

C2 asks a pure revision provider for a proposal, then authorizes and commits its own decision.

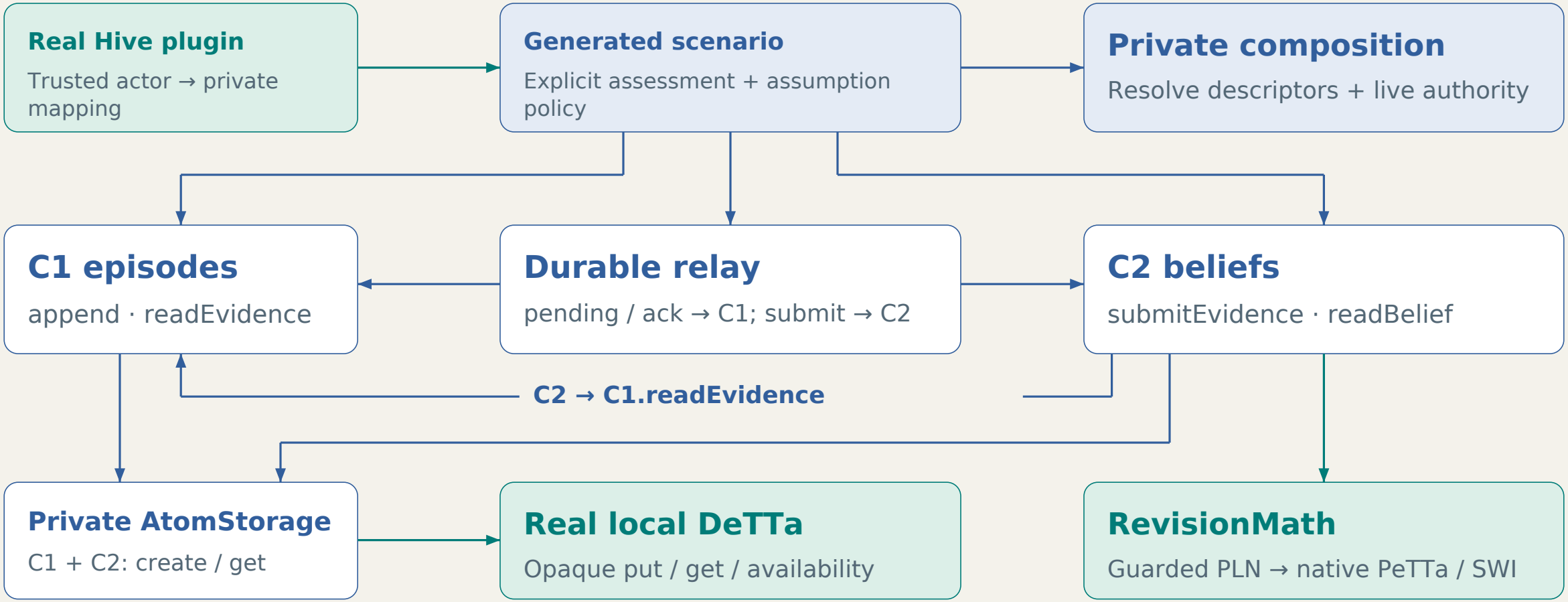
Control responsibilities: move work, preserve authority

The remaining six modules connect owners and report specific facts.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / STATUS
Messaging	Accept, deliver, acknowledge send · poll · ack	Inbox + deduplication · IMPLEMENTED
Execution	Run one named operation submit · status · result	Jobs + completion outbox · IMPLEMENTED
Registry	Describe configured providers descriptors	Static configuration · IMPLEMENTED
Resolver	Match required semantics resolve	No mutable state · IMPLEMENTED
Consensus / finality	Submit approved commitment; observe submit · observe	Signed intents · HISTORICAL M1
Authorization	Check current caller + scope + method authorize / live policy checks	Operator policy · LOCAL FIXTURE

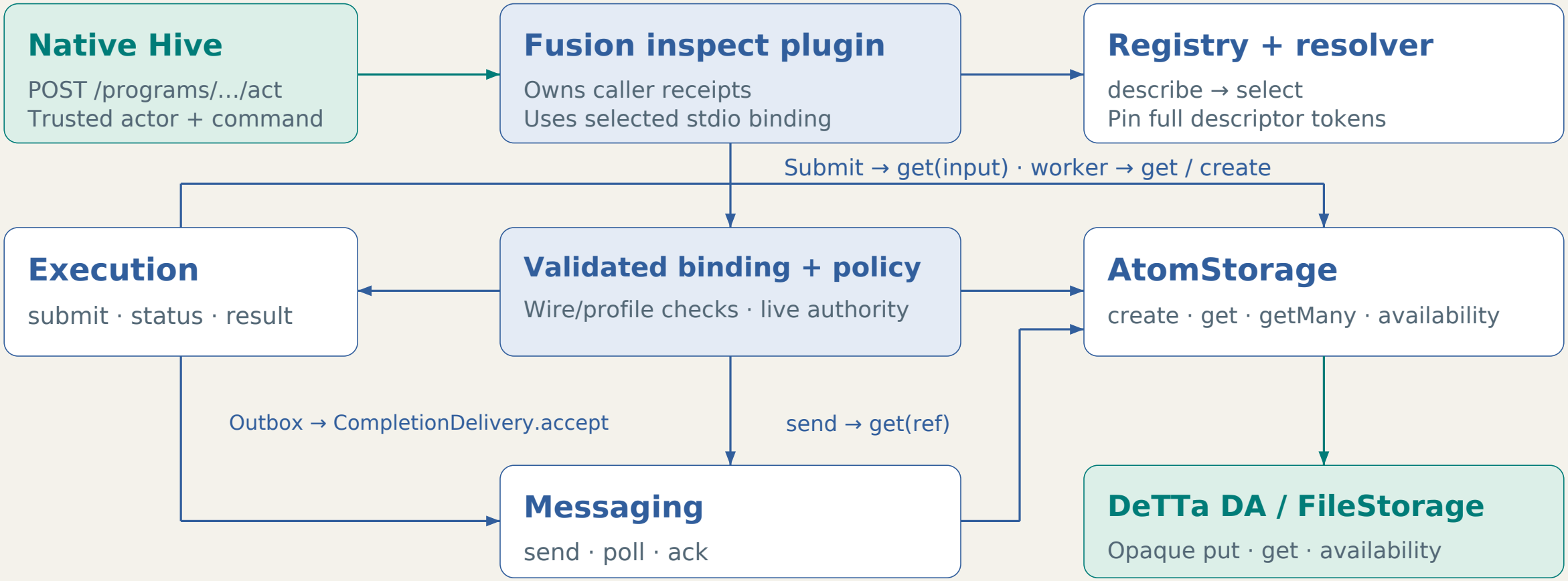
Function dependencies: the exercised M2 path

Solid arrows point from caller to provider. Each effect stays behind its declared port.



Function dependencies: the preserved M1.2 path

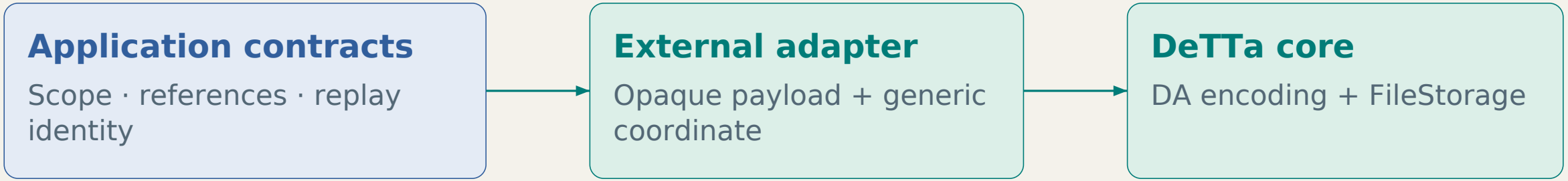
Solid arrows = exercised calls. Every provider effect crosses an existing validated interface.



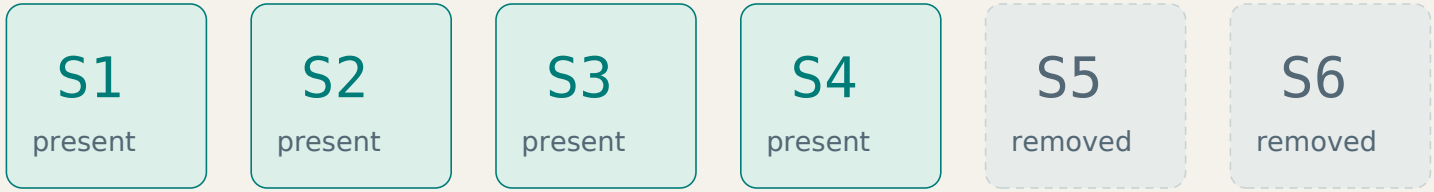
Optional finality uses a separate caller + approved digest + signer; M1.2 does not invoke it.

The DA boundary remains application-neutral

The external adapter translates application references into generic coordinates and opaque bytes.



4 data shares + 2 parity shares



Observed reconstruction

Cached payload + two shares removed: exact bytes recovered. A third missing share is unavailable.

The DA core does not interpret Hive identities, truth values, attention balances or job state.

Replace the provider; keep the consumer’s contract

Interface, implementation, transport and placement remain independent choices.

ATOMSTORAGE IMPLEMENTATION	IN-PROCESS BINDING	PRIVATE STDIO BINDING
Real DeTTa libraries	PASS · independent workflow	PASS · independent workflow
Atomic-file fixture	PASS · independent workflow	PASS · independent workflow

M2 C1 works without cognition

An independent, hostile port wrapper rejects access to provider internals or foreign SQL.

Compatibility is explicit

Select exact profiles, operations and descriptor tokens. Pin native source and binary identity.

Observed replacement uses fresh stores. Existing accepted state is not migrated.

Private history stays in a separate local composition

The authenticated actor selects an operator-configured scope; request data cannot nominate authority.

ONE PRINCIPAL · PRIVATE-LOCAL-ONLY

C1 owner

Events + head + delivery intents

C2 owner

Snapshots + attempts + decisions

Private AtomStorage

Separate payloads + metadata

Relay owner

Exact handoff requests + progress

Public Hive receipt

Generated counts + public command ID. Private lineage and content hashes stay local.

Shared services

No private mailbox, execution or chain port is constructed. Overlapping private/shared roots are rejected.

Privacy evidence covers the tested routes under a trusted host/operator with intact local files.

Contract gaps are resolved before adding behavior

Local v2 contracts state identity, ordering, numerical meaning and replay explicitly.

QUESTION	EXPLICIT CHOICE	CONSEQUENCE
What identifies an episode?	Stable scope + ID; separate mutable head	History grows without changing identity
What is a new occurrence?	Occurrence ID distinct from bytes and key	Same bytes can describe distinct events
What counts as evidence?	Explicit assessment + admitted provenance	A report alone has no truth value
What claim is revised?	Exact predicate, ordered arguments, polarity, context	No hidden semantic or negation conversion
What did submit do?	Admission outcome + numerical outcome	Accepted evidence may stay an alternative
What is retried?	Full semantic input + pinned runtime selection	Same key cannot acquire new meaning

C1: one occurrence, one position in the story

Immutable event atoms sit behind a stable episode identity and a committed sequence.

EpisodeRef = scope + ID

1 · Report

No numerical claim

2 · Assessment

Explicit strength / confidence

3 · Copy

Inherits original roots

4 · Contrary

Kept in the history

Same occurrence + same body

Return the original receipt, even under a new request key. The head does not advance.

Same occurrence + changed meaning

Conflict on body or provenance changes. Distinct real occurrences may share identical bytes.

C1 commits event + evidence intent together; AtomStorage has already committed independently.

C2: admitting evidence and changing belief are separate

New observations remain visible even when they cannot change the current aggregate.

Evidence submission

C1 evidence reference or an existing C2 support; exact claim and context.

Admission decision

ADMITTED or DUPLICATE
Track occurrence and provenance.

Numerical decision

Initialize, revise, no change, retain alternative, or model-inapplicable.

Aggregate with exact coverage

Every numerical result names the contributing leaves, origins, premises, formula and assumption.

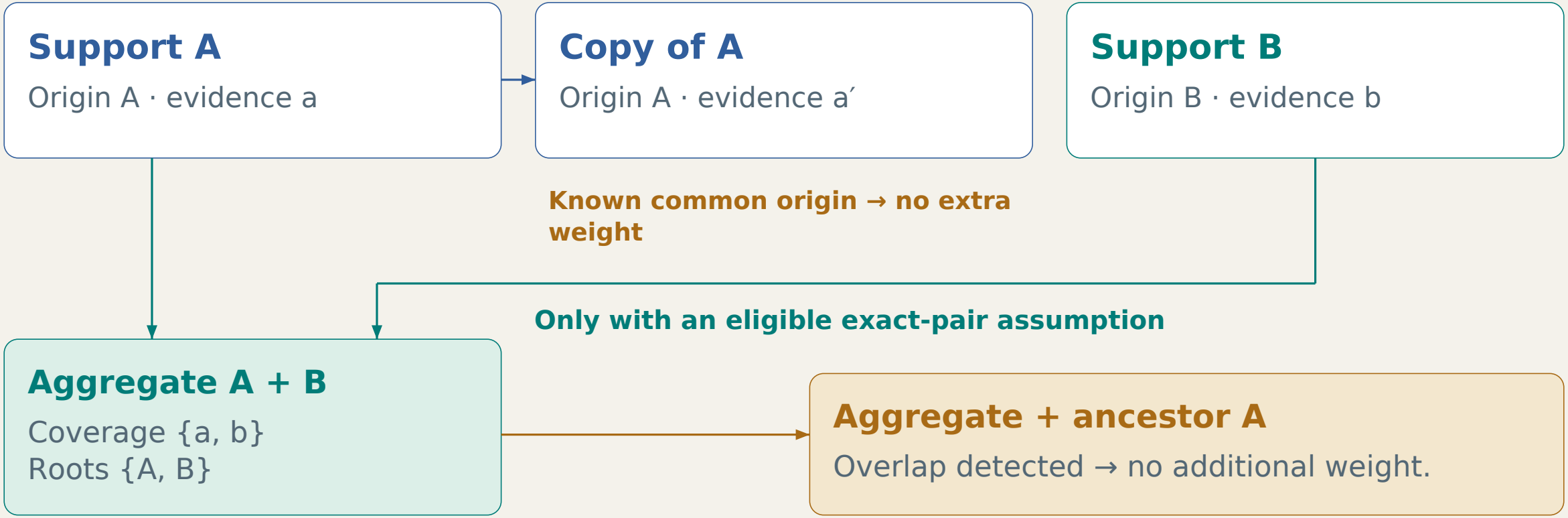
Unaggregated alternatives

Copied or contrary observations remain in the snapshot. Later eligible assumptions can combine existing supports.

One saved truth value is not a summary of every observation. Read its coverage and alternatives.

Lineage blocks double counting, even after aggregation

An independence declaration cannot override known overlapping roots or evidence.



New IDs, aliases or request keys do not manufacture independence.

Native revision reuses the guarded implementation

A narrow revision provider binds the upstream guards, formula and native runtime explicitly.

C2 owner
Exact support snapshots

→

PLNAdapter.revise
Unchanged Reachability guards

→

Truth_Revision
Pinned PLN source

→

PeTTa / SWI
Actual native evaluation

Explicit runtime rebaseline

Historical SWI 10.0.1 remains unchanged. This gate selects SWI 10.1.14 under a new revision-only profile.

Fail closed on drift

Source hashes, model, runtime profile and built binary digest are bound to durable attempts.

No replacement production mathematics, applyRule, PLN.Derive or PLN.Query was introduced.

Independent goldens check the actual native result

Pairs are explicit assessments of the same claim, with permitted independence and disjoint lineage.

INPUT PAIR · (STRENGTH, CONFIDENCE)	EXPECTED NATIVE RESULT	MEANING
(0.75, 0.50) + (0.25, 0.50)	(0.50, 2/3)	Equal weight balances the assessments
(0.75, 0.50) + (0.25, 0.75)	(0.375, 0.80)	Higher confidence shifts the result
Same unequal pair, reversed	(0.375, 0.80)	Order does not change this revision

Domain: finite values only

0 ≤ strength ≤ 1; 0 ≤ confidence < 1

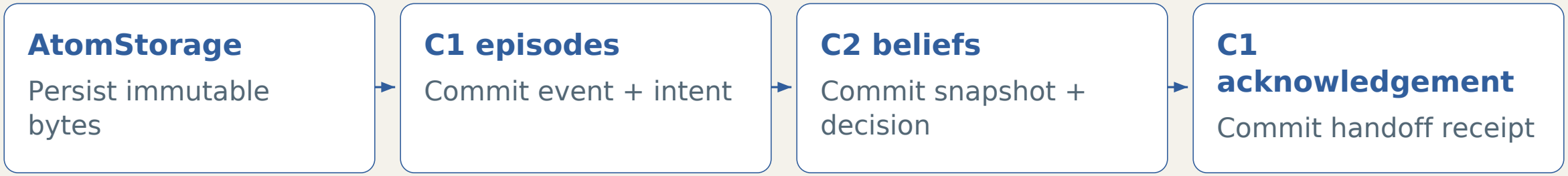
No silent clamping

Invalid inputs and zero total weight are refused.

Confidence belongs to the selected model; it is not a guarantee that the real-world claim is true.

Four crash gaps across independent commits

Restart recovers intent and decisions. Repeated calculation does not add another logical contribution.



KILLED AT THIS BOUNDARY	AFTER A NEW PROCESS STARTS
1 · Payload persisted; no C1 commit	Replay recovers the atom; an orphan is not an event.
2 · C1 intent committed; no C2 decision	The relay resubmits its exact durable request.
3 · Formula returned; no C2 commit	Calculation may repeat; CAS and replay guard the effect.
4 · C2 committed; no C1 acknowledgement	Recover the same decision, then acknowledge separately.

All four boundaries also run through the actual native Hive caller’s restart path.

Authority, replay and compatibility stay live

Saved success is evidence of an earlier decision, not permission to bypass current checks.

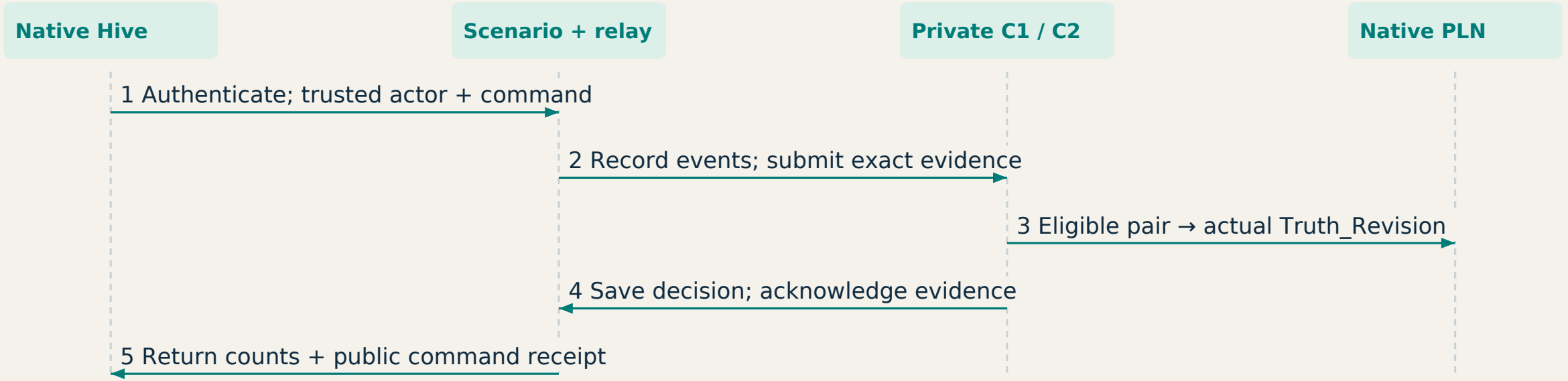
ATTEMPT OR INTERRUPTION	REQUIRED BEHAVIOR	PROTECTED BOUNDARY
Supply actor identity in Hive params	Native HTTP 400 identity_in_request	Trusted actor context
Revoke access after saved success	UNAUTHORIZED on replay	Current scope and method authority
Revoke authority during formula work	Recheck before C2 commit	No unauthorized state change
Change input under the same key	CONFLICT	Full semantic fingerprint
Change descriptor / source / runtime	Fail selected compatibility checks	No weaker fallback provider
Transient owner contention or timeout	Keep exact intent pending; no ack	Recoverability without extra evidence

A genuine CAS conflict needs a new key with the reconciled revision; transient retries keep the original key.

One actual Hive episode reaches private C1 and C2

Same verified Hive source and native Programs entrypoint; a sibling plugin hosts the generated scenario.

```
POST /api/programs/fusion-episode/act
```



7 native Hive tests passed: real app, authentication, Supervisor, PeTTa and plugin loader.

The recorded story keeps copies and disagreement

Generated scenario, real local DeTTa and native PLN; the CLI record is separate from the real Hive gate.

STORY EVENT	NUMERICAL OUTCOME	WHAT REMAINS VISIBLE
Report without an assessment	model-inapplicable	Original report; no invented truth value
Explicit assessment (0.75, 0.50)	initialized	One eligible support initializes the aggregate
Copy with the same origin	retained-alternative	Copy retained; no extra aggregate weight
Contrary assessment (0.25, 0.50)	retained-alternative	Disagreement retained without an assumption

4 events · 2 unaggregated alternatives

Duplicate appends do not advance the episode.

Separate eligible pair → (0.5, 2/3)

Native revision; ancestor re-entry adds no weight.

Reopening preserves the exact snapshots and decisions. The native Hive route exercises the same scenario.

M1.2 remains a working, separate integration

Private cognition is added alongside the original storage, messaging and execution path.

Original native operation

fusion-inspect still stores bytes, shares a ref, delegates inspect_atom, recovers completion and acknowledges it.

New private operation

fusion-episode records private events and revises eligible evidence. It constructs no mailbox, execution or chain port.

PRESERVED SOURCE BOUNDARY	FINAL REGRESSION
M1 wire schemas, registry/resolver, AtomStorage, DA adapter	Byte-identical to the requested M1.2 baseline
Mailbox, execution, original Hive plugin and driver	Original behavior preserved
Original native Hive integration tests	13 passed; test source unchanged

Upstream guarded sources are unchanged. Application semantics remain above opaque DA.

Real providers and fixtures answer different questions

A real runtime can execute generated data. Neither label cancels the other.

COMPONENT	CLASSIFICATION	WHAT THE EVIDENCE SUPPORTS
Hive app / Supervisor / PeTTa	REAL NATIVE RUNTIME	Actual authenticated dispatch and caller restart
DeTTa DA + FileStorage	REAL LOCAL PROVIDER	Opaque persistence, reconstruction, owner recovery
Guarded PLN + Truth_Revision	REAL NATIVE REVISION	Pinned guards + actual formula goldens
Generated data / origin assumptions	FIXTURES	Deterministic scenario, not empirical truth
File provider / RecordingRuntime / injections	FIXTURES	Independent contracts and refusal behavior
Fusion policy / transport / failpoints	TEST SUPPORT	Local grants and controlled failure placement

Native Hive authentication is real. Downstream Fusion authorization is a local development policy.

Executed suites cover distinct layers of the claim

Counts overlap: the 51 original fixtures and 16 M2 fixtures are included in the 106-case matrix.

SUITE	CASES	CLASSIFICATION / OBSERVED USE
Full conformance matrix	106	Original 90 + 16 M2 fixtures; separate 51-fixture run
Clean DeTTa libraries	62 + 38	Real DA and FileStorage
Hive baseline + original native integration	34 + 13	Baseline fixture driver / real native caller
M2 isolated fixtures + upstream guard fixtures	16 + 8	Real-provider paths absent / RecordingRuntime
M2 native revision	12	Actual goldens; malformed-output injections labelled
M2 recovery / privacy + real Hive	6 + 7	Actual SIGKILL, CAS race, native authenticated route

All final commands returned 0. Native suites reported zero skipped tests. No new runtime tests run for this deck.

M2 gates: contracts, memory and native revision

All required M2 gates passed for the supported local profile.

GATE	RESULT	EVIDENCE OBLIGATION MET
BASELINE	PASS	Original behavior and native integration preserved
CONTRACTS	PASS	Closed local inputs / outputs, profiles, bounds and fingerprints
C1	PASS	Real-backed ordered history, replay, provenance and intents
C1_INDEPENDENT	PASS	Independent AtomStorage; cognition absent; declared ports only
C2_GUARDS	PASS	Unchanged upstream guards + hostile-input refusals
C2_NATIVE	PASS	Actual native equal / unequal / reversed-pair goldens

M2 gates: privacy, recovery, Hive and reproducibility

No M2 prerequisite remains blocked. Optional or excluded work retains its own status.

GATE	RESULT	OBSERVED BOUNDARY
PRIVACY	PASS	Tested private routing; cross-principal / scope and revocation
RECOVERY	PASS	Four real kill boundaries, stable decisions, two-process CAS
HIVE_M2	PASS	Seven actual authenticated native caller tests
REPRO	PASS	Fresh providers + clean Fusion run; download caches warm
SCOPE	PASS	C1 + C2 only; M1.2 and upstream guarded sources preserved
CHAIN	NOT_RUN	Outside M2; no chain action or fallback

Historical exact chain-source reproduction remains BLOCKED; it is not a missing M2 prerequisite.

Reproducibility binds source, build and execution

The final code is tested from a clean detached checkout; native inputs are independently pinned.

1 · Fetch exact sources

Hive + DeTTa via authorized SSH; Reachability, PLN, PeTTa and SWI at full commits.

2 · Build fresh providers

New native objects, venv, Janus and DeTTa target. Download caches may be warm.

3 · Verify exact Fusion

Clean d9e7577 checkout. All supported gates; exact commands, hashes and logs retained.

Observed build: 337.325 seconds

A build observation on this host; no throughput or latency claim.

Evidence snapshot: f0cee24

Documentation/evidence descendant with the same tested implementation.

A reproducible source/build recipe does not promise bit-identical binaries across hosts.

The guarantees stop at the tested boundary

Useful evidence is specific about the assumptions that made it possible.

OBSERVED GUARANTEE	LIMIT OR UNRUN EXTENSION
Duplicate-safe occurrences and numerical contributions	Physical formula evaluation may repeat after a crash
Retained contrary observations and exact coverage	No proof of source truth; undisclosed copying is undetected
Private local placement and tested route isolation	Trusted host/operator; no encryption or admin secrecy claim
Independent owner commits; process restart recovery	Power loss and distributed failover NOT_RUN
Fresh-store substitution; explicit model/runtime profiles	No live state migration or universal-model claim
Fresh pinned build; selected native Hive tests	Empty-cache build, historical SWI 10.0.1, full Hive suite NOT_RUN

Stopped at M2: no ECAN, inference control, general rule kernel, chain action or live migration.

Remember the event.
Keep the disagreement.
Justify the update.

Each state owner commits its part. Each retry keeps its meaning.

Real Hive · real local DeTTa · actual guarded PLN revision

M2 COMPLETE · C1 + C2 ONLY

Implementation map: each owner has a small surface

Composition selects providers. Domain owners consume declared ports.

FILES / AREA	RESPONSIBILITY	STATE OR EXCLUSION
memory_types.py · memory_contracts.py · ports.py	Local C1/C2 DTOs, protocols and validation	No provider construction
private.py	Private placement, descriptors, live authority	Only composition chooses concrete owners
episodes.py	C1 occurrences, head, replay and intents	episodes.sqlite
beliefs.py	C2 snapshots and semantic attempts	beliefs.sqlite
evidence_relay.py	Bounded exact evidence handoff	relay.sqlite; no foreign SQL
adapters/pln_revision.py	Pinned guards + revision-only native runtime	Pure proposal; no truth setter
integration/m2_scenario.py · episode_program/	Generated policy + sibling native Hive plugin	Private owners; minimal public receipt

Exact sources include the explicit runtime rebaseline

Prefixes are navigation aids. Full SHAs and formula/binary digests are embedded in these notes.

INPUT	COMMIT / VERSION	ROLE
Fusion tested / evidence	d9e75773c349 / f0cee248294a	Clean run / same-code evidence descendant
Schematics → actual Hive	4af00db4e395 → 34b0fb32ab38	Intended native source and Programs route
DeTTa	272888f264dd	Real clean local DA / FileStorage
Reachability	dd56f3f053ed	Unchanged guarded PLNAdapter.revise
PLN	4405956947c4	Actual Truth_Revision formula
PeTTa / SWI source	ae66fa8e41dc / 69775434c822	Native runtime at explicit source pins
SWI / Python / Janus	10.1.14 / 3.12.12 / 1.5.3	Narrow rebaseline; historical 10.0.1 unchanged

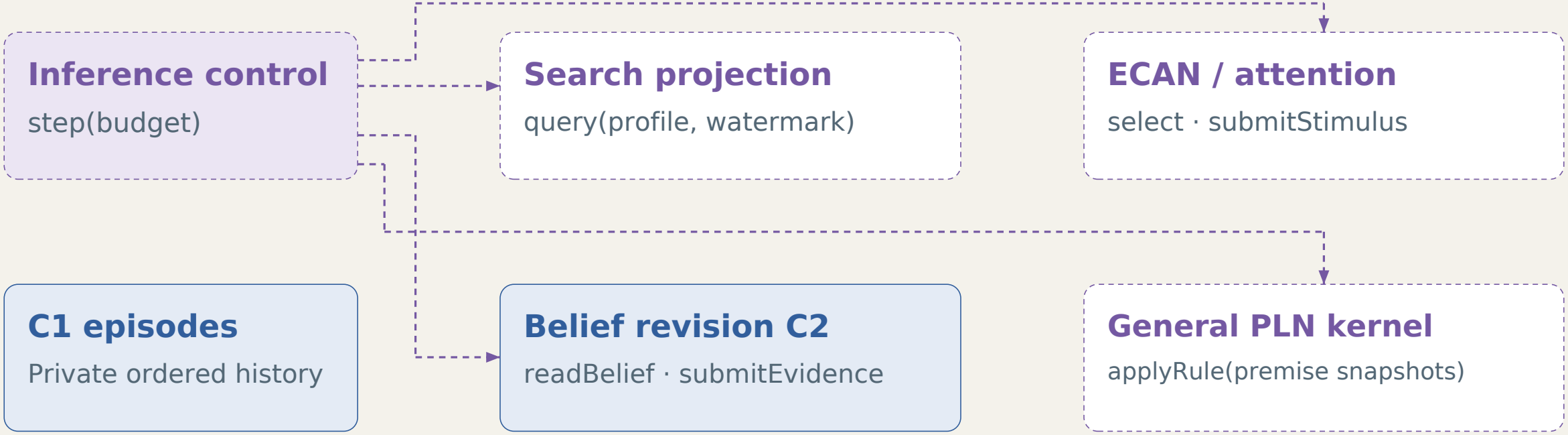
Profiles and bounds are part of compatibility

A matching method name alone does not establish substitutability.

SELECTION	EXACT PROFILE OR LIMIT	IMPLICATION
C1 / C2 local interface	2.0.0; implementation 0.2.0	No new remote cognitive wire routes
C1 / C2 semantics	episode-local-sequence/2 · pln-revision-private/1	Bound identity, ordering and numerical decisions
Numerical / runtime profiles	trueagi-pln-stv-finite-k1/v1 fusion-petta-swi-10.1.14-revision/1	Explicit model + native selection
Private atom / event bytes	65,536 / 16,384 bytes	Immutable bounded records
Episode / provenance / batch	256 events; 8 provenance + 8 causation; page 16	No unbounded query surface
Belief / snapshot / relay	32 observations; 64 supports; 262,144 bytes 3 CAS retries/drain; 32 attempts/delivery	Retained history; no GC or total disk quota

Future cognition remains a set of review obligations

Dashed arrows are specified future calls. The M2 owners in blue are already implemented.

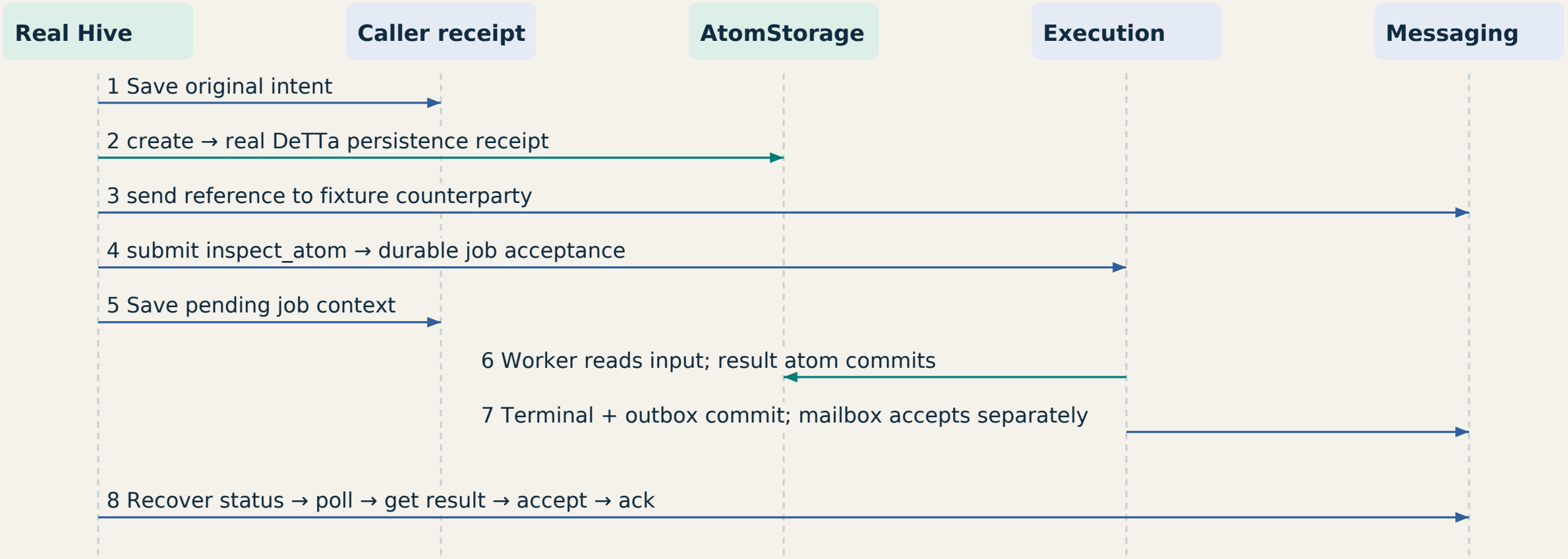


Attention: budget, allocation, decay, tie-breaking. Rules: model, premises, golden tests.

Control: bounded work and feedback. Search: profile and freshness watermark.

One native operation crosses the existing interfaces

The caller stores intent; each provider commits only its own state.



Historical M1: one result reached an isolated chain

This is the earlier protocol-peer artifact. No M1.2 or M2 Hive result was submitted.

PENDING
Signed deploy accepted

→

INCLUDED
Block observed; effects pending

→

FINALIZED
Deploy effects reported finalized

M1 recorded observation

Block height	**3**
Execution error	**false**
Rejection count	**0**
Execution cost	**351 phlo**

Historical commitment digest

```
f9236451ff785dc58315251cf864a3d1
0938412551cf45f448055061bc953ea1
```

One local validator; direct node observations.
Not a consensus proof or production test.

M2 CHAIN = NOT_RUN. Historical source reproduction requires exact F1R3 changes or a separate rebaseline.

Reproduce the supported M2 gates

Run from the evidence revision or a reviewed descendant. See the runbook for Linux and source access.

BUILD FRESH PINNED PROVIDERS

```
uv run --python 3.11 --locked python \
scripts/build_m2.py artifacts/m2-providers
```

VERIFY BASELINE, C1, C2, PRIVACY, RECOVERY AND REAL HIVE

```
uv run --python 3.11 --locked python \
scripts/verify_m2.py artifacts/m2-providers/environment.json
```

Read: manifest.json → final/verification.json → per-suite logs → final/demo.json

Fixture success, native provider success, blocked historical inputs and unrun extensions remain separate.

Retry decisions preserve meaning across owners

Separate acceptance, numerical effect and acknowledgement before choosing a recovery action.

OBSERVATION	MEANING	NEXT ACTION
Lost response / process death	The owner may already have committed	Replay exact input, key and pinned selection
NOT_AVAILABLE / transient contention	No final decision yet	Keep intent pending; retry unchanged
CONFLICT: stale expected revision	Another owner-local update won	Read current snapshot; new key for revised intent
CONFLICT: same key, changed input	Replay identity no longer matches	Do not overwrite the original meaning
UNAUTHORIZED / profile mismatch	Access or selected semantics fail	Correct authority / compatibility; no fallback
Decision durable; ack missing	C2 effect complete, C1 handoff pending	Recover same decision; acknowledge full receipt

Trusted local acknowledgement binds the complete decision; it is not a portable cryptographic proof.